

- 1 The Earth may be considered as a uniform sphere of radius  $6.37 \times 10^6 \text{ m}$ .

Cambridge is at a point on the Earth's surface that has a latitude of  $52.2^\circ$  north of the Equator, as shown in Fig. 1.1.

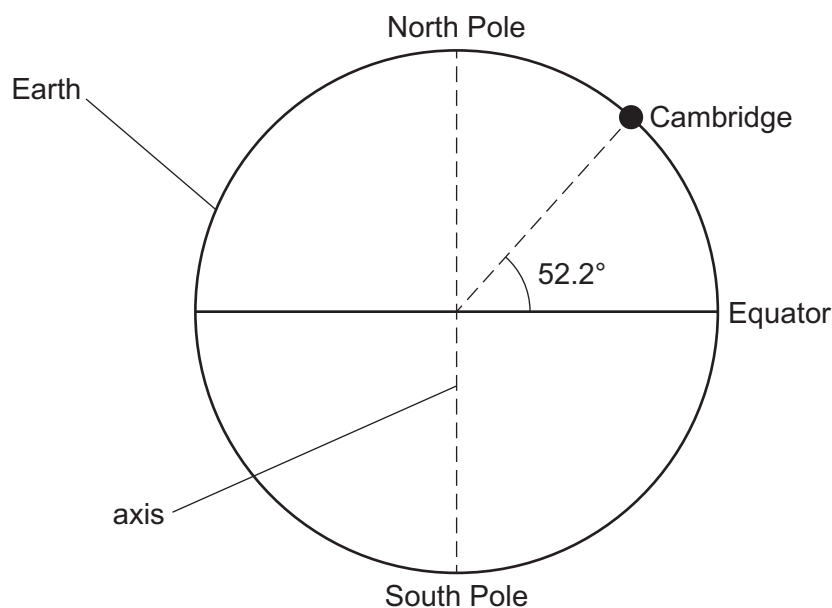


Fig. 1.1

As the Earth spins on its axis, Cambridge moves in a circle that is parallel to the Equator but with a smaller radius.

- (a) (i) Show that the radius of the circle around which Cambridge moves is  $3.90 \times 10^6 \text{ m}$ .

[1]

- (ii) Calculate the speed at which Cambridge moves around the circle.

speed = .....  $\text{ms}^{-1}$  [3]

